

Kasra Farsoudi

farsoudi@gmail.com | www.linkedin.com/in/farsoudi | www.github.com/farsoudi | farsoudi.com

EDUCATION

University of Southern California

Los Angeles, CA

B.S. in Computer Engineering & Computer Science

Aug. 2024 – Dec. 2026

Relevant Coursework: Professional C++, Machine Learning, MOS VLSI Circuit Design, Parallel Design, Data Structures, Embedded Systems, Digital Circuits, Distributed IoT Systems, Algorithms & Theory of Computing, Software Development, Operating Systems, Computer Architecture, Networking

EXPERIENCE

Full-Stack Software Engineer

Jul. 2023 – Present

Brake Performance

Los Angeles, CA

- Implemented Apple Pay end-to-end on an e-commerce platform with frontend payment session handling on Next.js, and backend charge processing with PHP Symfony. Accounts for 20% of overall sales.
- Built a fraud-detection engine using MySQL that indexes new orders against flagged orders.
- Built a cron for AWS Linux that ingests shipment data from UPS & USPS and updates order status.
- Designed a full-stack sales report system with API Platform and React Admin, giving management dynamic day/month/year revenue breakdowns by order, part, and product across the entire catalog.
- Designed a reusable transactional email system for wholesale approvals, shipment errors, and fraud alerts, with fail-safe logging so a failed email never blocks an order.

Junior Software Engineer (*Lead Frontend Engineer*)

October 2022 – July 2023

I-Safe

Carlsbad, CA

- Developed the front-end logic for handling 25+ Rest API endpoints in TypeScript on React Native.
- Built the Redux state management system, including user data storage and synchronization to React lifecycle.
- Integrated Redux-Saga middleware for RESTful API calls and pipelining asynchronous data to client-side storage.
- Designed and implemented the entire login and authorization workflow.
- Integrated secure token-based authentication with refresh logic in JavaScript.
- Built custom React Native components for document signing portable across Android & iOS.

PROJECTS

Self-Hosted AI Agent | *Ollama, Fedora Linux, Docker, PostgreSQL, RAG, n8n*

August 2026

- Architected a fully private AI assistant stack on a headless RTX 3090 Fedora server, accessed over Tailscale
- Ran local LLM inference with Ollama, serving Qwen 3.8 27B
- Configured a LibreChat (Docker) chat UI wired to Ollama endpoint, with custom MCP tooling
- Integrated MongoDB, Meilisearch, and pgvector backends for conversations, search, and RAG
- Designed a cross-device persistent memory service in FastAPI + SQLite, using FTS5 plus 768-dim embeddings and reciprocal-rank-fusion hybrid search
- Built an n8n + Postgres automation layer: an LLM email agent classifying Gmail and extracting deadlines, writing approved items to Google Calendar/Tasks

Paid LLM Inference Gateway | *Go, x402, PostgreSQL, Ollama*

September 2026

- Built a Go gateway that fronts a self-hosted Ollama instance with metered, pay-as-you-go inference
- Funded prepaid USDC balances via the x402 HTTP 402 payment protocol, integrating Coinbase's facilitator on Base to verify and settle on-chain payments
- Designed an off-chain Postgres ledger to credit balances and record auditable usage transactions

FPGA Street Fighter | *Verilog, Xilinx Vivado*

April 2025

- Synthesized a 2-player fighting game on Nexys A7 FPGA hardware with custom hard-wired breadboard controllers
- Designed the core game engine in Verilog: collision detection, combat resolution, and win-condition logic
- Built the player state machine and a reusable timer module driving movement, jump arcs, and shield mechanics

TECHNICAL SKILLS

Languages: Go, C/C++, JavaScript, SQL (MySQL & Postgres), PHP, Verilog, CUDA, Python

Frameworks: React / React Native, Node.js, Symfony, API Platform, ESP32 / ESP-IDF, MongoDB

Developer Tools: Git, Docker, Tailscale, n8n, Neovim, Ollama, AWS

Linux Distros: Arch (Personal Distro), Amazon Linux, CentOS, Fedora, Ubuntu, Raspbian